

Syllabus:- Limit

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If $[x]$ is greatest integer $\leq x$ then $\lim_{x \rightarrow 0^+} \left[\frac{1}{x^2} \right] \ln(\cos x) =$
 (1) $-1/2$ (2) $-1/3$ (3) 0 (4) does not exist
- If a sequence $\langle a_n \rangle$ is such that $a_1 = 1$, $a_{n+1} = \frac{2+3a_n}{1+2a_n}$ and $\lim_{n \rightarrow \infty} a_n$ exists then $a_n =$
 (1) $\cos 36^\circ$ (2) $\sin 18^\circ$ (3) $2 \cos 36^\circ$ (4) $\sin 36^\circ$
- $\lim_{x \rightarrow 2} \frac{\sqrt{1+\sqrt{2+x}} - \sqrt{3}}{x-2} =$
 (1) $\frac{1}{8\sqrt{3}}$ (2) $\frac{-1}{4\sqrt{3}}$ (3) 0 (4) $\frac{1}{4\sqrt{3}}$
- $\lim_{x \rightarrow 0^+} (\log(\sin^3 x) - \log(x^4 + ex^3)) =$
 (1) 0 (2) 1 (3) -1 (4) does not exist
- If f is a real valued differentiable function such that $|f(x) - f(y)| \leq (x - y)^2$ where $x, y \in \mathbb{R}$ and $f(0) = 0$ then $f(1) =$
 (1) -1 (2) 0 (3) 2 (4) 1
- If $f(x) = \begin{cases} \frac{\sin[x]}{[x]}, & [x] \neq 0 \\ 0, & [x] = 0 \end{cases}$ where $[\cdot] = \text{GIF}$ then $\lim_{x \rightarrow 0} f(x) =$
 (1) 1 (2) 0 (3) -1 (4) limit does not exist
- The value of $\lim_{x \rightarrow 0} \frac{1}{x} \left[\tan^{-1} \left(\frac{x+1}{2x+1} \right) - \frac{\pi}{4} \right]$ is :
 (1) $-\frac{1}{2}$ (2) 1 (3) 2 (4) 0
- $\lim_{x \rightarrow 0} \frac{\log(\sin 7x + \cos 7x)}{\sin 3x}$ equals :
 (1) $\frac{1}{3} \log 7$ (2) $\frac{7}{3}$ (3) $\frac{14}{3}$ (4) $\frac{1}{3}$

9. $\lim_{x \rightarrow 0} x^2 \left(1 + 2 + 3 + \dots + \left\lceil \frac{1}{|x|} \right\rceil \right)$ where $[.]$ denotes greatest integer function, is equal to
- (1) 0 (2) $\frac{1}{2}$ (3) 2 (4) Limit does not exist
10. If $f(x)$ is differentiable and strictly increasing function then the value of $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is
- (1) 1 (2) 0 (3) -1 (4) 2
11. If $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 3^{2n}}{5^n + 2^n + 3^{2n+3}}$ is equal to
- (1) 5 (2) 3 (3) 1 (4) None of these
12. $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin x \cos x}$ is equal to
- (1) $\frac{2}{5}$ (2) $\frac{3}{5}$ (3) $\frac{3}{2}$ (4) $\frac{3}{4}$
13. $\lim_{n \rightarrow \infty} \left(\frac{1}{1-n^2} + \frac{2}{1-n^2} + \dots + \frac{n}{1-n^2} \right)$ is equal to:
- (1) 0 (2) $-\frac{1}{2}$ (3) $\frac{1}{2}$ (4) None of these
14. Let $f(x) = \frac{2\sqrt{2} - (\cos(x) + \sin(x))^3}{1 - \sin(2x)}$ then $\lim_{x \rightarrow \frac{\pi}{4}} f(x)$ is equal to
- (1) $\frac{1}{\sqrt{2}}$ (2) $\sqrt{2}$ (3) 1 (4) $\frac{3}{\sqrt{2}}$
15. If $f(x) = \begin{cases} x^2 + bx; & 0 \leq x \leq 1 \\ 3 - ax^2; & 1 < x \leq 2 \end{cases}$
- such that $\lim_{x \rightarrow 1} f(x) = 4$, then ordered pair (a, b) is given as-
- (1) (1, 5) (2) (1, 7) (3) (-1, 3) (4) None of these
16. $\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\frac{\pi}{2} - \theta}{2 \cot \theta}$ is equal to
- (1) 0 (2) -1 (3) 1 (4) ∞
17. $\lim_{x \rightarrow 0} \frac{(2^{\sin x} - 1)[\ln(1 + \sin 2x)]}{x \tan^{-1} x}$ is equal to
- (1) $\ln 2$ (2) $2 \ln 2$ (3) $(\ln 2)^2$ (4) 0
18. The value of $\lim_{x \rightarrow \pi/2} \frac{\sin x - (\sin x)^{\sin x}}{1 - \sin x + \ln \sin x}$ is-
- (1) 0 (2) 1 (3) 2 (4) None of these
19. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(1) = 3$ and $f'(1) = 6$, Then, $\lim_{x \rightarrow 0} \left\{ \frac{f(1+x)}{f(1)} \right\}^{1/x}$ is equal to
- (1) 1 (2) $e^{1/2}$ (3) e^2 (4) e^3
20. If $\lim_{x \rightarrow \infty} \{ (\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - cx^2 + 3x - d}) \}$ is finite, then the value of a is
- (1) 3 (2) 5 (3) 2 (4) any real number

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. if $\lim_{x \rightarrow 0} \frac{\sin^3 x}{ax e^x - b \ln(1+x) + cx e^{-x}} = \frac{3}{2}$ then $2b + a + c =$

22. If α is interior angle of a regular octagon, then $\lim_{\theta \rightarrow \alpha^+} \frac{\tan \theta - 1}{[\sin \theta + \cos \theta]}$ is equal to

[Note : $[k]$ denotes greatest integer less then or equal to k]

23. If $f(x) = \begin{cases} \frac{\sqrt{g(x)} - 1}{\sqrt{x} - 1}; & x \neq 1 \\ 1; & x = 1 \end{cases}$ and $g'(1) = 2, g(1) = 1$, then $\lim_{x \rightarrow 1} f(x)$ is equal to

24. If $\lim_{x \rightarrow 0} \frac{3 \tan^{-1} x + 3 \tan x - x^5 - 6x}{3x^n}$ is a finite number then the greatest value of n is

25. $\lim_{x \rightarrow \frac{\pi}{2}^-} [1 + (\cos x)^{\cos x}]^2$ is equal to

Answer Key_DPP_09

1.	(1)	2.	(3)	3.	(1)	4.	(3)	5.	(2)	6.	(4)	7.	(1)
8.	(2)	9.	(2)	10.	(3)	11.	(4)	12.	(3)	13.	(2)	14.	(4)
15.	(3)	16.	(3)	17.	(2)	18.	(3)	19.	(3)	20.	(3)	21.	(12)
22.	(02)	23.	(2)	24.	(7)	25.	(4)						